

Saturation Temperature & Pressure Lab

ENME 601 Thermodynamics

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Synopsis

We measured the relationship between saturation temperature and absolute pressure for water using the Armfield saturation–pressure apparatus and compared the results with steam table values. Gauge pressure readings were converted to absolute pressure with the measured atmospheric pressure, and the corresponding theoretical saturation temperatures were obtained by interpolation from the steam tables. The data was fitted with the power law

$$T_{\text{sat}} = A P_{\text{abs}}^n, \quad (1)$$

for both the experimental series and the steam table series. The experimental and theoretical dataset aligned closely across the tested range (from 100 to 165 °C): experimental $A = 30.738$, $n = 0.2569$; steam table $A = 30.745$, $n = 0.257$. The mean temperature error is $\sim 0.030\%$. The small remaining differences, most visible at the highest pressures, are consistent with thermal lag and minor systematic effects (see Figure 1).

1 Aim

To measure and model the dependence of water’s saturation temperature on absolute pressure, then verify the experimental results by cross-checking with published steam-table data.

2 Theory

At saturation, liquid water and its vapour coexist in equilibrium at a single temperature that depends on pressure. Plotting this equilibrium condition produces a smooth saturation line in T – P space over the specified range. Over a limited range, the relation can be approximated by the power law in Eq. (1), which provides a simple model for comparison to steam-table data.

3 Equipment

Armfield saturation-pressure apparatus (boiler with sight-glass and viewing port); power/control console; pressure gauge (P_1); platinum resistance thermometer (T_1); filling-point, isolating, and drain valves; barometer for P_{atm} ; purified/de-ionised water.

4 Method

1. Ensure that the isolating and drain valves are closed; turn off console power. Fill with de-ionised water to mid sight-glass; leave the filling-point valve open.
2. Heat to vigorous boiling to purge air; maintain steady steam flow until temperature stabilises, then close the filling-point valve to create a closed, constant-volume saturated mixture.
3. Step the heater power. After each step, reduce power and wait for T_1 to stabilise to mitigate thermal lag (sensor and fluid at the same temperature).
4. Record steady readings of T_{exp} (from T_1) and gauge pressure P_1 for 7 points up to ~ 600.000 kPa
5. Convert gauge to absolute pressure:

$$P_{\text{abs}} = P_{\text{atm}} + P_1. \quad (2)$$

6. Interpolate the saturation temperature from the steam tables, T_{th} , at each P_{abs} .
7. Compute percentage error:

$$\text{Error \%} = \frac{T_{\text{exp}} - T_{\text{th}}}{T_{\text{th}}} \times 100. \quad (3)$$

8. Fit the power model (Eq. 1) to both series using Excel graphing capabilities.

5 Results

5.1 Atmospheric Pressure (lab day)

Measured P_{atm} (barometer): 101.600 kPa.

TABLE 1: SATURATION LAB RESULTS

Gauge pressure $\rightarrow$ absolute pressure; measured temperature T_{exp} ; resistance R (raw) and corrected resistance R_{corr} .

P_1 (kPa)	$P_{1,\text{abs}}$ (kPa)	T_{exp} ($^{\circ}\text{C}$)	R (Ω)	R_{corr} (Ω)
0	101.6	100	138.7	138.71
100	201.6	120	145.0	146.04
200	301.6	134	148.6	151.17
300	401.6	146	151.5	155.17
400	501.6	152	153.8	157.91
500	601.6	160	155.7	160.71
600	701.6	162	157.2	162.13

TABLE 3: EXPERIMENTAL AND INTERPOLATED THEORETICAL RESULTS AND CORRESPONDING ERROR PERCENTAGE

Interpolated T_{th} from steam tables at each $P_{1,\text{abs}}$; error computed by Eq. (3).

$P_{1,\text{abs}}$ (kPa)	T_{exp} ($^{\circ}\text{C}$)	T_{th} ($^{\circ}\text{C}$)	Error (%)
101.6	100	100.0397	-0.04
201.6	118	120.4506	-0.37
301.6	126	133.7522	0.19
401.6	134	143.7473	1.57
501.6	140	151.9462	0.04
601.6	146	158.9308	0.67
701.6	150	165.0396	-1.87

TABLE 2: POWER-LAW FIT PARAMETERS FOR $T = AP_{\text{abs}}^n$

Power law applied to T_{exp} vs P_{abs} & Steam table T_{th} vs P_{abs} over the measured range.

	A	n
Experimental data	30.738	0.2569
Steam-table data	30.745	0.257

5.2 Power-Law Fits and Usefulness

The fitted power laws are listed in Table 2. The two exponents are very similar around ($n \approx 0.257$), and A differs only in the second decimal place, indicating a very close match in scaling between the experimental and steam-table saturation data.

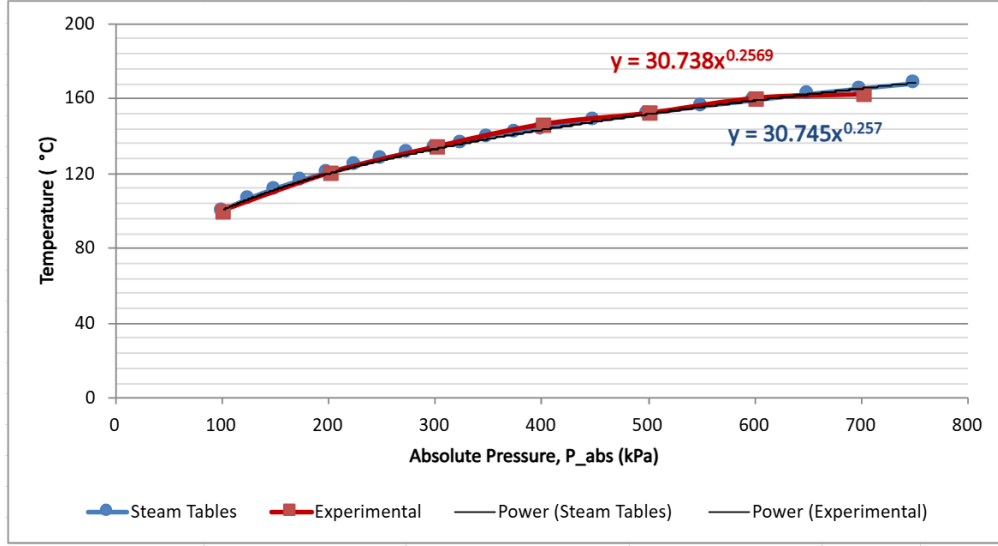


Figure 1: Temperature vs absolute pressure with power-law trendlines

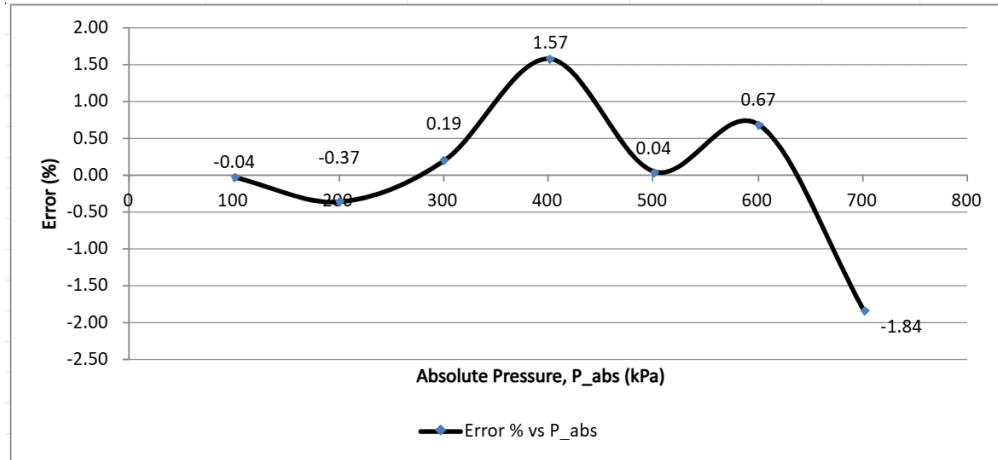


Figure 2: Percentage error vs absolute pressure

6 Discussion

Comparison of curves. The experimental T - P_{abs} curve overlays the steam-table saturation line closely across 100.000 °C–165.000 °C (Figure 1). Differences are small and visually consistent, but some larger differences at 400.000 kPa & 700.000 kPa are notable as seen in (Figure 3).

Accuracy and fitted equation. The fits are tight. From Table 2 we obtain $A = 30.738$ and $n = 0.2569$ for the experimental set, and $A = 30.745$ and $n = 0.257$ for the steam table. Thus n matches to three significant figures and A differs by only 0.007. One power law describes this range well and provides a quick way to compare the two sets. For exact values or for pressures outside the tested range, the steam tables are a reliable source.

Error percentage (trend and average). Most points fall within Error $\lesssim 0.700\%$, with two larger deviations at 401.600 kPa and 701.600 kPa (+1.57% and -1.84%). The average error over all seven points is about 0.030%, indicating no meaningful systematic bias.

Steam vs water temperature; thermal lag and heat loss. At true equilibrium, the liquid and the steam sit at the same saturation temperature. In the system, the probe takes a moment to catch up, and some heat leaks along the probe and fittings, so the reading can dip when things are changing—especially at higher pressure where changes are sharper. The two bigger deviations likely show up where this lag had affect. The stabilisation waits kept the bias small.

Possible sources of error. Instrument calibration (gauge and thermometer), barometer reading for P_{atm} , interpolation calculation error between table entries, minor leakage at elevated pressure, and reading/notetaking errors. The corrected resistance values in Table 1 display that measuring tools have their limitations, and there will be some error inherited from rounding in this step. Outside of experimental and documentation; The temperature deviations could be explained by thermal lag and small systematic effects caused within the system.

7 Conclusion

The experiment confirmed the expected smooth saturation relationship between temperature and absolute pressure for water. Across the measured range, the experimental and steam-table curves line up closely. The fitted power-law exponents are essentially the same and the average error is $\sim 0.030\%$. The two bigger deviations at higher pressures are consistent with thermal lag and small systematic effects in the setup. The power-law model shows that experimental data is most useful when compared, and our supplied thermodynamic tables using interpolation techniques are a quality source for precise values.